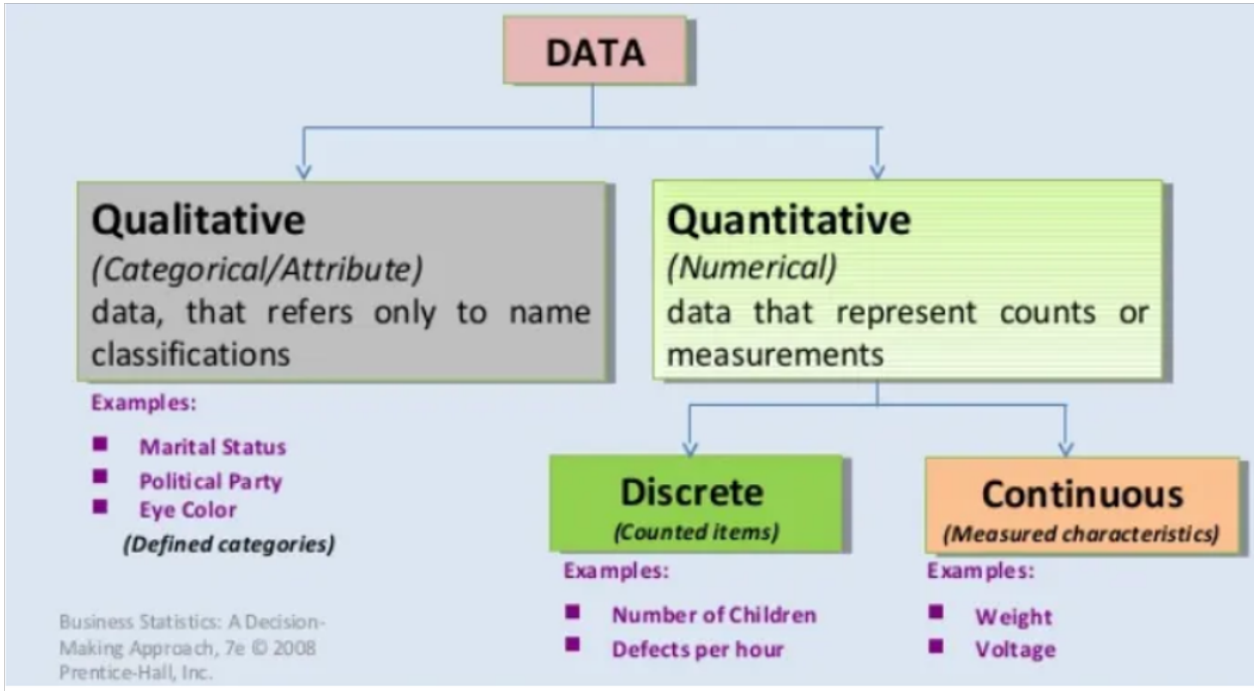
BECS 22623
Implementation of Numerical Methods using Appropriate software

Functional Interpolation & Extrapolation

Department of Mathematics
Faculty of Science
University of Kelaniya

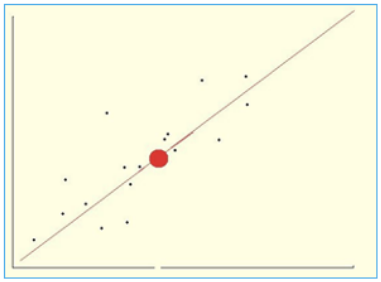
What is Data?

In computing, **data** are information that has been translated into a form that is efficient for movement or processing.



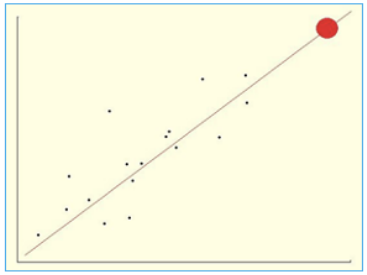
Definition of Interpolation & Definition of Extrapolation

Interpolation is a technique for adding new data points within a range of a set of known data points.



Interpolation

Extrapolation is a process for estimating dependent variable values for independent variable values outside of the fitting data domain.



Extrapolation

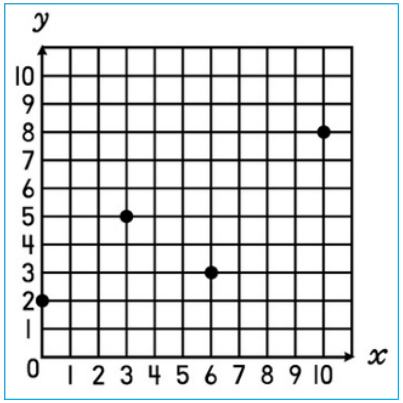
Applicability & Possibilities

These techniques are important for:

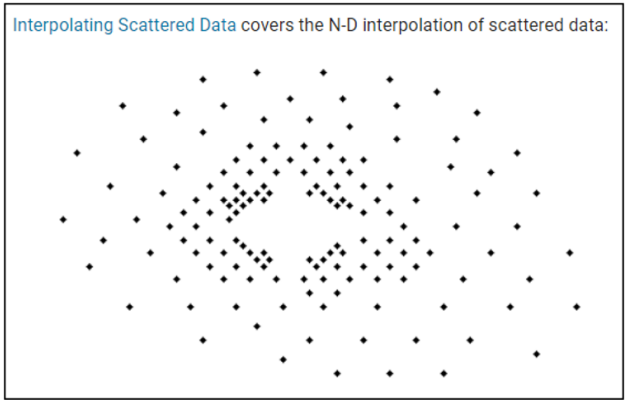
- fill-in missing data,
- smooth existing data,
- make predictions etc.

In MATLAB, techniques is designed mainly for

data points on a grid



scattered data points



Interpolation-Grid Data in MATLAB

- A wide range of grid data interpolation techniques are available in MATLAB.

Functions

1-D and Gridded Interpolation

`interp1`	1-D data interpolation (table lookup)
`interp2`	Interpolation for 2-D gridded data in meshgrid format
`interp3`	Interpolation for 3-D gridded data in meshgrid format
`interpn`	Interpolation for 1-D, 2-D, 3-D, and N-D gridded data in ndgrid format
`griddedInterpolant`	Gridded data interpolation
`pchip`	Piecewise Cubic Hermite Interpolating Polynomial (PCHIP)
`makima`	Modified Akima piecewise cubic Hermite interpolation *(Since R2019b)*
`spline`	Cubic spline data interpolation
`ppval`	Evaluate piecewise polynomial

Interpolation with *Interp1/2, interpn*

`interp1`	1-D data interpolation (table lookup)
`interp2`	Interpolation for 2-D gridded data in meshgrid format
`interp3`	Interpolation for 3-D gridded data in meshgrid format
`interpn`	Interpolation for 1-D, 2-D, 3-D, and N-D gridded data in ndgrid format

- $[X, Y] = meshgrid(x, y)$ - returns 2-D grid coordinates based on vectors x and y .
- X is a matrix where each row is a copy of x , and Y is a matrix where each column is a copy of y .
- $[X1, X2, X3, \ldots Xn] = ndgrid(x1, x2, x3, \ldots, xn)$- replicates the grid vectors x1,x2,x3.. to produce an n-dimensional full grid.

Interpolation with *Interp1/2, interpn*

$$vq = interp1(x, v, xq)$$
$$vq = interp1(x, v, xq, method)$$
$$vq = interpn(x, v, xq,' method')$$

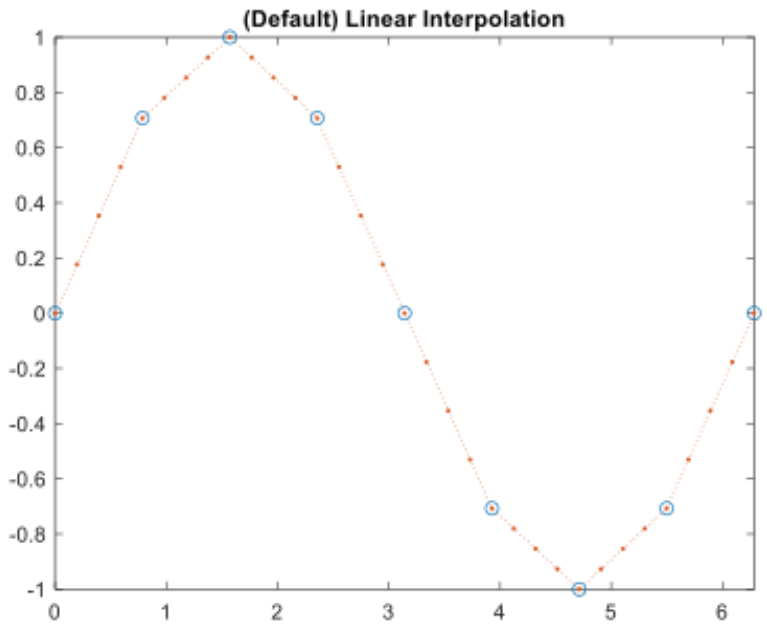
x	v	xq	vq	method
sample points	V(x) values	Query points	Interpolated values	Interpolation method
length(x) must equal length(v) or size(v,1)	specified as a vector, matrix of real or complex numbers	specified as a scalar, vector, matrix	Interpolated function values at Query points	• Linear • Nearest • Next • Previous • Cubic etc.
[1 2 3 4 5 6 7 8], 1:8	rand(1,10)	1:0.01:8		
(https://in.mathworks.com/help/matlab/ref/interp1.html#btwp6lt-1-xq)				

Methods

Method	Description	Continuity	Comments
`'linear'`	Linear interpolation. The interpolated value at a query point is based on linear interpolation of the values at neighboring grid points in each respective dimension. This is the default interpolation method.	C^0	• Requires at least 2 points • Requires more memory and computation time than nearest neighbor
`'nearest'`	Nearest neighbor interpolation. The interpolated value at a query point is the value at the nearest sample grid point.	Discontinuous	• Requires at least 2 points • Modest memory requirements • Fastest computation time
`'next'`	Next neighbor interpolation. The interpolated value at a query point is the value at the next sample grid point.	Discontinuous	• Requires at least 2 points • Same memory requirements and computation time as `'nearest'`
`'previous'`	Previous neighbor interpolation. The interpolated value at a query point is the value at the previous sample grid point.	Discontinuous	• Requires at least 2 points • Same memory requirements and computation time as `'nearest'`
`'pchip'`	Shape-preserving piecewise cubic interpolation. The interpolated value at a query point is based on a shape-preserving piecewise cubic interpolation of the values at neighboring grid points.	C^1	• Requires at least 4 points • Requires more memory and computation time than `'linear'`
`'cubic'`	Cubic convolution used in MATLAB® 5.	C^1	• Requires at least 3 points • Points must be uniformly spaced • This method falls back to `'spline'` interpolation for irregularly-spaced data • Similar memory requirements and computation time as `'pchip'`
`'v5cubic'`	Same as `'cubic'`.	C^1	

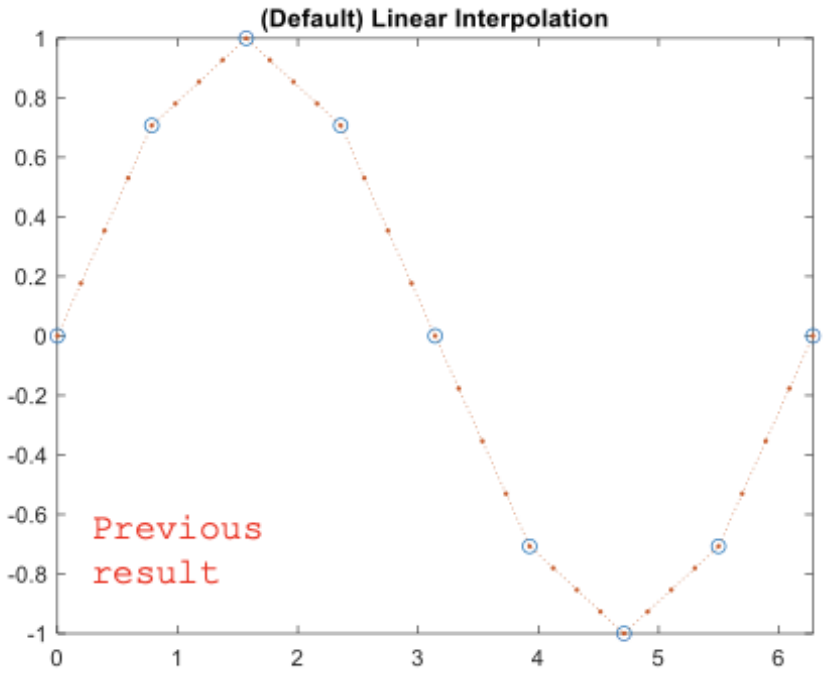
Example using Interp1, interpn

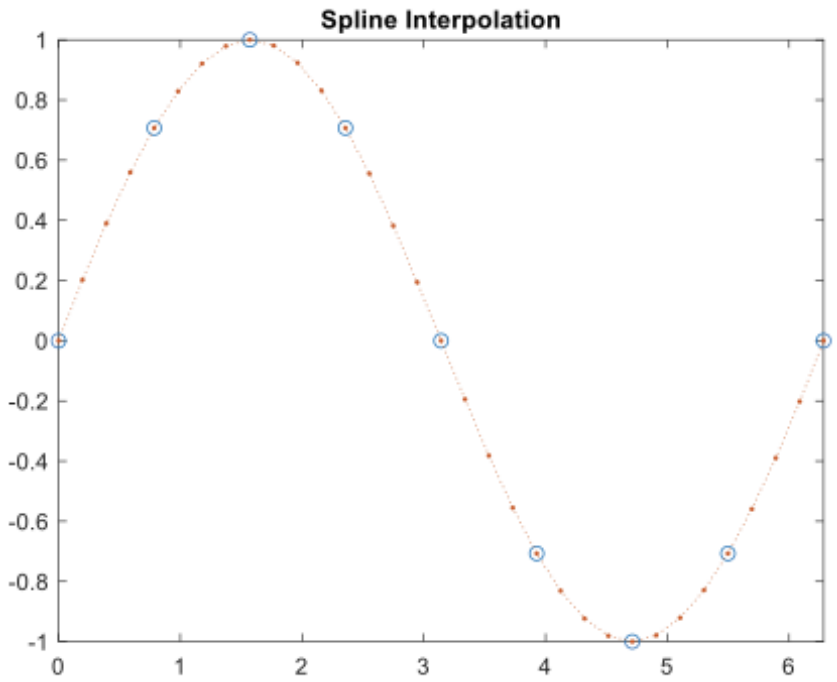
```
x = 0:pi/4:2*pi;
v = sin(x);
xq = 0:pi/16:2*pi;
vq1 = interp1(x,v,xq);

figure()
plot(x,v,'o',xq,vq1,':.');
xlim([0 2*pi]);
title('(Default) Linear
Interpolation');
```



Example using Interp1, interpn with **'spline'**

```
vq2 = interp1(x,v,xq,'spline');
figure()
plot(x,v,'o',xq,vq2,':.');
xlim([0 2*pi]);
title('Spline Interpolation');
```





Cubic Spline Interpolation

- A spline is a function made up of a piecewise series of polynomials, with different polynomials used to evaluate the function in different regions.
- A cubic spline is a spline function made up of cubic polynomials.
- Cubic spline interpolation is a numerical method used to approximate a smooth curve passing through or near a given set of data points.
- It is widely employed in applications such as computer graphics, engineering, and data fitting.

Cubic Spline

Definition 1

A cubic spline $S(x)$ through the data points $(x_1, y_1), \cdots (x_n, y_n)$ is a set of cubic polynomials of the form

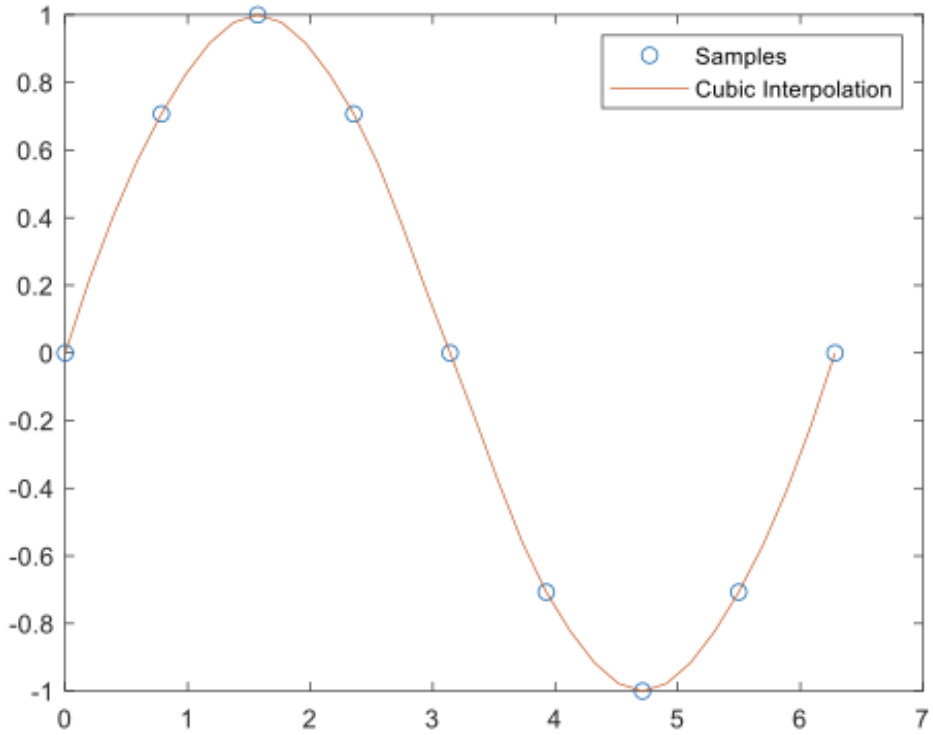
$$S_i(x) = y_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, x \in [x_i, x_{i+1}], i = 1, 2, \cdots n - 1$$

that satisfy the following three properties:

1. $S_i(x_i) = y_i$ and $S_i(x_{i+1}) = y_{i+1}, \ i = 1, \cdots n - 1.$
2. $S'_{i-1}(x_i) = S'_i(x_i), \ i = 2, \cdots, n - 1.$
3. $S''_{i-1}(x_i) = S''_i(x_i), \ i = 2, \cdots, n - 1.$

Example using Interp1, interpn with **'cubic'**

```
vq = interpn(x,v,xq,'cubic');
figure()
plot(x,v,'o',xq,vq,'-');
legend('Samples','Cubic Interpolation');
```



Remark: Consider the following statement:

```
vq = interpn(v);
```

This defines the grid as a set of points who's spacing is 1 and range is [1, size(V,i)] in the ith dimension.

Interpolation -Scattered Data

- We use ***ScatteredInterpolant*** to perform interpolation, extrapolation on a 2-D or 3-D data set of scattered data.
- As before first we create the interpolant F for the given data set.

$$F = ScatteredInterpolant(x, y, v)$$
$$F = ScatteredInterpolant(x, y, v, method)$$

- Then evaluate the F at interested query points (xq, yq) as

$$Vq = F(xq, yq)$$

Extrapolation with Matlab

$$Vq = interp1(x, v, xq, method, extrapolation),$$

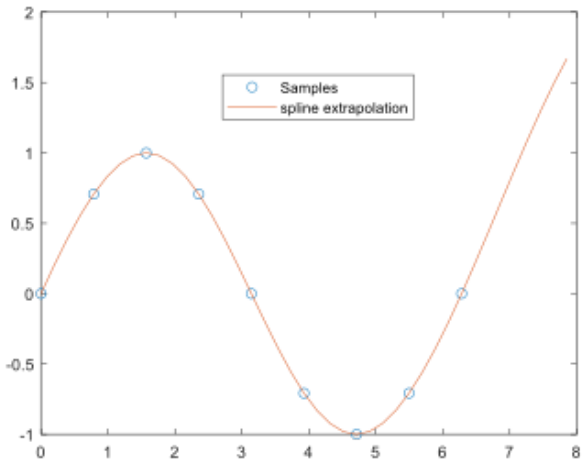
- Uses specifically for evaluating the points that lie outside the domain of x.
- 'extrap' is the method used for extrapolation.
- Extrapolation method can be 'nearest', 'linear', or 'none'.

$$F = griddedInterpolant(x, y, v, method, ExtrapolationMethod)$$
$$F = scatteredInterpolant(x, y, v, method, ExtrapolationMethod)$$

- Specifies both the interpolation and extrapolation methods

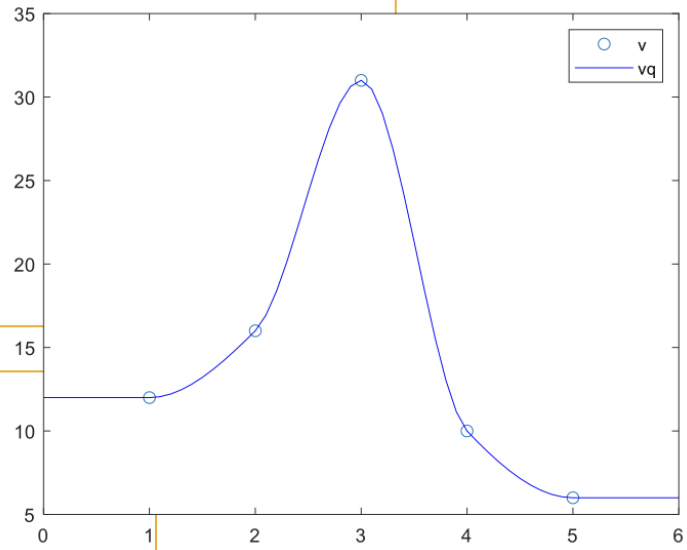
Example using **interp1**

```
clc, clear all, close all
x = 0:pi/4:2*pi;
v = sin(x);
xq = 0:pi/16:2.5*pi;
vq = interp1(x,v,xq,'spline','extrap');

figure()
plot(x,v,'o',xq,vq,'-');
legend('Samples','spline extrapolation');
```



Example using **griddedInterpolant**

```
clc, clear all, close all
x = [1 2 3 4 5];
v = [12 16 31 10 6];
F = griddedInterpolant(x,v,'pchip','nearest')

xq = 0:0.1:6;
vq = F(xq);
figure()
plot(x,v,'o',xq,vq,'-b');
legend ('v','vq')
```




```
 F= griddedInterpolant with properties:

            GridVectors: {[1 2 3 4 5]}
                 Values: [12 16 31 10 6]
                 Method: 'pchip'
   ExtrapolationMethod: 'nearest'
```

Department of Mathematics

Faculty of Science, University of Kelaniya.